Linear Algebra

→ Matrix

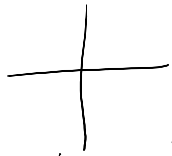
$\circ$ [ ] → pixels

RGB

→ Vector → $\vec{a}$

Physics [ ]

Math



2-D

3-D

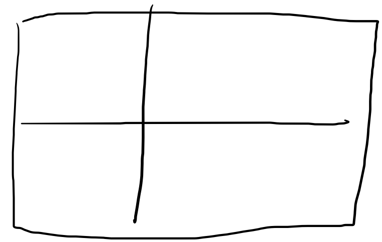
Unit vectors

2-D → $\hat{i}, \hat{j}$

3-D → $\hat{i}, \hat{j}, \hat{k}$

Basis →

Pair of → 2-D



Linear combination

$\vec{a}, \vec{b}$

$$\vec{y} = c_1 \vec{a} + c_2 \vec{b}$$

↳ Linear Comb.

Linear Com

=

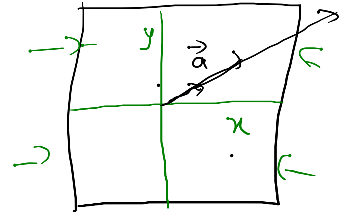
vector

$\vec{y} = c_1\vec{a} + c_2\vec{b} + c_3\vec{c} + c_4\vec{d} + \dots$

scalar | weight

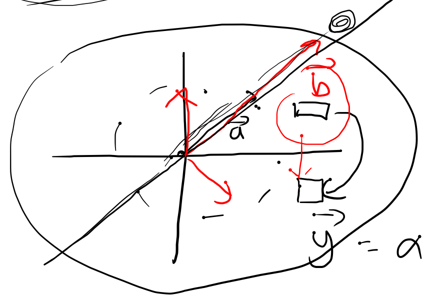
NR's

basis



For 2-d → a pair

$\vec{y} = \alpha_1 \vec{a}$



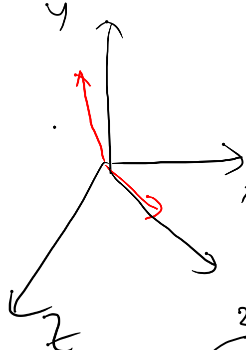
→ Space

Span : $\vec{a}, \vec{b}$

$y = \alpha_1 \vec{a} + \alpha_2 \vec{b}$

Span

$\vec{y} = \alpha_1 \vec{a} + \alpha_2 \vec{b}$



3 →

3 vectors

3 →

Plane

$y =$

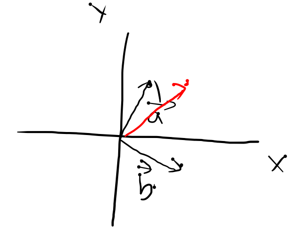
a pair → $\vec{a}$ $\vec{b}$

$\vec{y} = \alpha_1 \vec{a} + \alpha_2 \vec{b}$

Ex.

$$3 \to \left\{ \underset{\vec{a}}{\begin{pmatrix}1\\2\end{pmatrix}}, \underset{\vec{b}}{\begin{pmatrix}3\\-1\end{pmatrix}}, \underset{\vec{c}}{\begin{pmatrix}3\\2\end{pmatrix}} \right\}$$

Linearly dependent



Are they lin.

$\vec{c}$

$$\begin{bmatrix}3\\2\end{bmatrix} = \begin{bmatrix}\alpha_1\\2\alpha_1\end{bmatrix} + \begin{bmatrix}3\alpha_2\\-\alpha_2\end{bmatrix}$$

$$\begin{pmatrix}3\\2\end{pmatrix} \quad \begin{pmatrix}1\\2\end{pmatrix} \quad \begin{pmatrix}3\\-1\end{pmatrix}$$

$$\vec{c} = \alpha_1 \vec{a} + \alpha_2 \vec{b}$$

α_1, α_2

$$2 = \frac{18}{7} - \alpha_2$$

$$\begin{bmatrix}3\\2\end{bmatrix} = \alpha_1\begin{bmatrix}1\\2\end{bmatrix} + \alpha_2\begin{bmatrix}3\\-1\end{bmatrix}$$

$$3 = \alpha_1 + 3\alpha_2 \quad \text{— (1)}$$

$$\alpha_2 = 4/7$$

$$2 = 2\alpha_1 - \alpha_2 \quad \text{— (2)}$$

multiply (2) by 3 & add (1)

α_1 & α_2 exist ✓

$$3 = \alpha_1 + \cancel{3\alpha_2}$$
$$6 = 6\alpha_1 - \cancel{3\alpha_2}$$
$$9 = 7\alpha_1 \Rightarrow \alpha_1 = 9/7$$

$$\left\{ \begin{pmatrix}1\\2\\3\end{pmatrix} \quad \begin{pmatrix}1\\-1\\2\end{pmatrix} \quad \begin{pmatrix}2\\1\\5\end{pmatrix} \right\}$$

$$\vec{a} \qquad \vec{b} \qquad \vec{c}$$

$$2 = \alpha_1 + \alpha_2$$

$$1 = 2\alpha_1 - \alpha_2$$

$$5 = 3\alpha_1 + 2\alpha_2$$

$\alpha_1 = 1$ and $\alpha_2 = 1$

Linearly dependent

$$\vec{c} = \alpha_1 \vec{a} + \alpha_2 \vec{b}$$

α_1 & α_2 exist

$$\begin{bmatrix}2\\1\\5\end{bmatrix} = \alpha_1 \begin{bmatrix}1\\2\\3\end{bmatrix} + \alpha_2 \begin{bmatrix}1\\-1\\2\end{bmatrix}$$

$$\Rightarrow \begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}1\\-1\\2\end{pmatrix}, \begin{pmatrix}4\\1\\3\end{pmatrix}$$
$$\vec{a} \qquad \vec{b} \qquad \vec{c}$$

$$\vec{c} = \alpha_1 \vec{a} + \alpha_2 \vec{b}$$

$$\begin{bmatrix}4\\1\\3\end{bmatrix} = \alpha_1 \begin{bmatrix}1\\2\\3\end{bmatrix} + \alpha_2 \begin{bmatrix}1\\-1\\2\end{bmatrix}$$

$$\begin{bmatrix}4\\1\\3\end{bmatrix} = \begin{bmatrix}\alpha_1\\2\alpha_1\\3\alpha_1\end{bmatrix} + \begin{bmatrix}\alpha_2\\-\alpha_2\\2\alpha_2\end{bmatrix}$$

$$\left\{\begin{aligned} 4 &= \alpha_1 + \alpha_2 \quad - (1) \\ 1 &= 2\alpha_1 - \alpha_2 \quad - (2) \\ \Rightarrow 3 &= 3\alpha_1 + 2\alpha_2 \quad - (3) \end{aligned}\right\} \checkmark$$

$$4 = \alpha_1 + \cancel{\alpha_2}$$
$$1 = 2\alpha_1 - \cancel{\alpha_2}$$
$$5 = 3\alpha_1$$

$$\Rightarrow \alpha_1 = 5/3$$

$$\alpha_2 = 7/3$$

$$4 = \frac{5}{3} + \alpha_2$$

$$\Rightarrow \alpha_2 = 4 - \frac{5}{3} = \frac{12-5}{3} = 7/3$$

$$3 = \beta \cdot \frac{5}{3} + 2 \cdot \frac{7}{3}$$

$$3 = 5 + \frac{14}{3}$$

$$3 \neq 19/3$$

Linearly independent